Assignment - 4

Section - A

Ans 1 : SRAM vs DRAM → SRAM uses flip flops to store ~~faster~~ data, making it faster, expensive and power consuming. It is used in cache memory.
DRAM uses capacitors to store data, needs refreshing, slower but cheaper. It is used as main memory.

Ans 2 : Programs tend to access a small portion of memory repeatedly.
Type
- → Temporal locality - recently accessed data is reused
- → Spatial locality - nearby locations are accessed

Ans 3 : Cache memory stores frequently used data | instructions closer to CPU. It reduces avg memory access time and improves performance.

Ans 4 : Cache Indexing
- → Direct mapped cache
- → Set associative cache
- → Fully associative cache

Ans 5 : Larger cache block size increases spatial locality [fewer misses] but increases miss penalty and may load unnecessary data.

Ans 6 : Associativity is the number of locations where a memory block can be placed in cache. Higher associativity reduces conflict misses but increases hardware cost.

Ans 7 : Larger cache size reduces miss rate and improves performance but increases cost, power and access latency

Ans 8 :

(i) Write-through : Data is written to both cache and main memory at the same time

(ii) Write-Back : Data is written to cache first and main memory is updated only when the block is replaced.

Ans 9 : AMAT = Hit time + (Miss rate × Miss penalty)

Hit time → time to access cache

Miss rate → probability of miss

Miss penalty → time to fetch block from main memory

Ans 10 : LRU block is replaced first. ~~This~~ The block which has not been used for the longest time is removed.

Ans 11 : Memory is divided into multiple modules and accessed simultaneously in a round robin fashion. It increases memory bandwidth and improves performance.

Ans 12 : Tracks are concentric circular paths on disk surface. Sectors are subdivisions of tracks used to store data blocks.

Ans 13: Flash memory is a non-volatile memory that retains data even when power is off. Used in SSDs, pen drives, memory cards. Faster than magnetic disks and has no moving parts.

Ans 14: I/O devices share the same address space as memory. Devices are accessed using normal memory instructions.

Ans 15: I/O devices have a ~~st~~ separate address space. They are accessed using special I/O instructions.

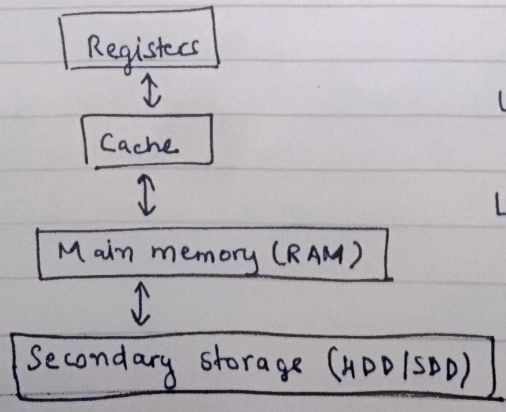
Ans 16: CPU continuously checks the status of I/O device and controls the data transfer. CPU is busy waiting.

Ans 17: Device interrupts the CPU when it is ready. CPU can perform other tasks meanwhile. Better CPU utilization

Ans 18: DMA allows I/O devices to transfer data directly to/from memory without CPU involvement. It increases efficiency.

Section B

Ans 1:



Data flows b/w different levels.
Upper levels are faster and smaller.
Lower levels are slower but larger in capacity.

Ans 2

Address		
Tag (t bits)	Index (i bits)	Offset (b bits)

Cache lines

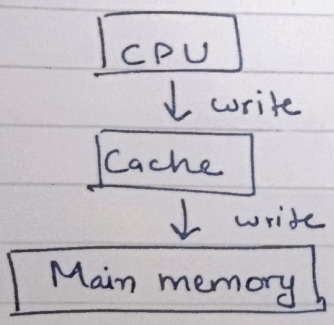
	Valid	Tag	Data Block (2^b bytes)
0			
1			
2			
⋮			
$2^i - 1$			

Each memory block maps to only one cache line using index.

Ans 3 :

i) Write - through

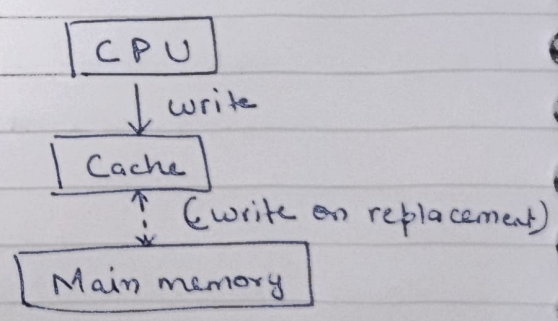
→



Each write updates both cache and main memory

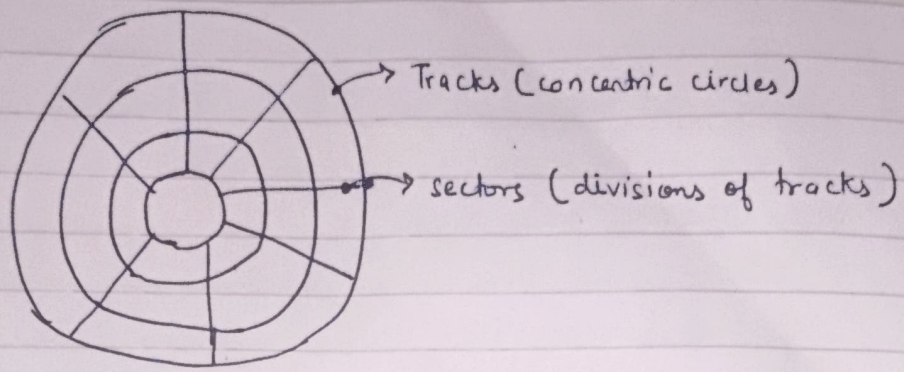
ii) Write - Back

→



Write updates only cache
Main memory updated when block is replacement.

Ans 4 :



Section - C

Ans 1 : Temporal locality, because the same location is accessed again and again

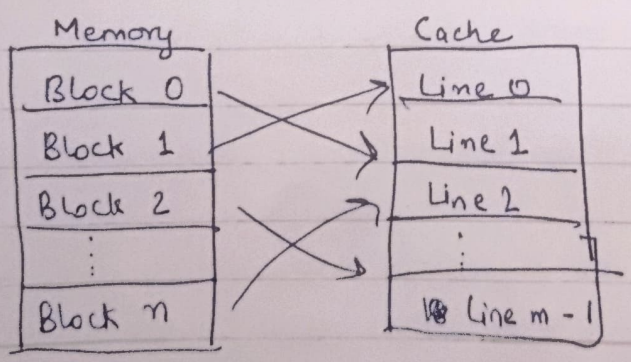
Ans 2 : Larger blocks increase miss penalty (more data to be transferred) and may load unnecessary data (cache pollution)

Ans 3 : On every write, data is written to both cache and main memory

Ans 4 : Direct Memory Access (DMA)

Section - D

Ans 1 :



Mapping : memory block is placed in cache line using Index. Tag is used in Index correctness.

Ans 2 :

$$\boxed{\text{AMAT} = \text{Hit time} + (\text{Miss rate} \times \text{Miss penalty})}$$

Factors affecting performance
- → Lower hit time
- → Lower miss rate
- → Lower miss penalty.

These reduces AMAT and improve performance

Ans 3 :

Programmed I/O	Interrupt I/O	DMA
→ CPU checks device continuously	Device interrupts CPU when ready	DMA controller transfers data to/from memory directly
→ CPU is busy waiting	Better CPU utilization	Best efficiency

Section - E

Ans 1 : Two assumptions while explaining cache operation
- → Fixed block size is assumed
- → Single level cache is considered

Ans 2 : Increases memory traffic because every write updates main memory.

Ans 3 : Cache mapping is most difficult because it involves address breakdown (tag, index, offset) and mapping logic which is conceptual and needs practice.